

- 2 The orca, *Orcinus orca*, has the largest distribution of all aquatic mammals and is found in nearly all seas and oceans. Orca are social mammals that usually live in groups. These groups can vary in size.

Fig. 2.1 shows an orca.



Fig. 2.1

There are a number of distinct types of orca. These distinct types of orca are classified as members of the same species. However, there is evidence that sympatric speciation is occurring.

- (a) (i) There are two distinct types of orca in the Northeast Atlantic Ocean: Type 1 and Type 2. Type 1 orca feed mainly on fish. Type 2 orca feed mainly on aquatic mammals, such as seals.

Fig. 2.2 shows the locations in the Northeast Atlantic Ocean where Type 1 orca and Type 2 orca have been observed. Orca do **not** occur only in these areas and some groups of orca travel great distances.

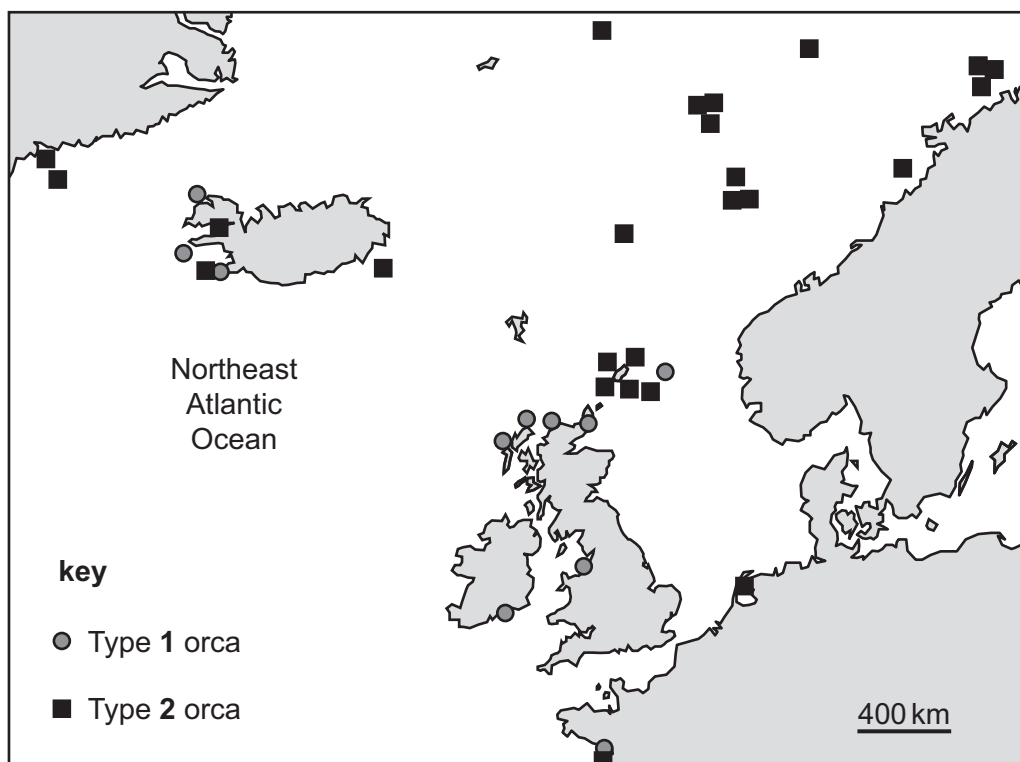


Fig. 2.2



With reference to Fig. 2.2, explain why the type of speciation that is occurring in the orca is described as sympatric speciation.

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(ii) Suggest examples of behavioural separation that would contribute to sympatric speciation of Type 1 orca and Type 2 orca.

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- (b) In the Southern Ocean, which surrounds Antarctica, there are three distinct types of orca: Type **B**, Type **C** and Type **D**.

Fig. 2.3 shows the locations around Antarctica where Type **B** orca, Type **C** orca and Type **D** orca have been observed.

- Type **B** orca and Type **C** orca are mainly seen near the coastline of Antarctica (inshore).
- Type **D** orca are mainly seen in the Southern Ocean further away from the coastline of Antarctica (offshore).

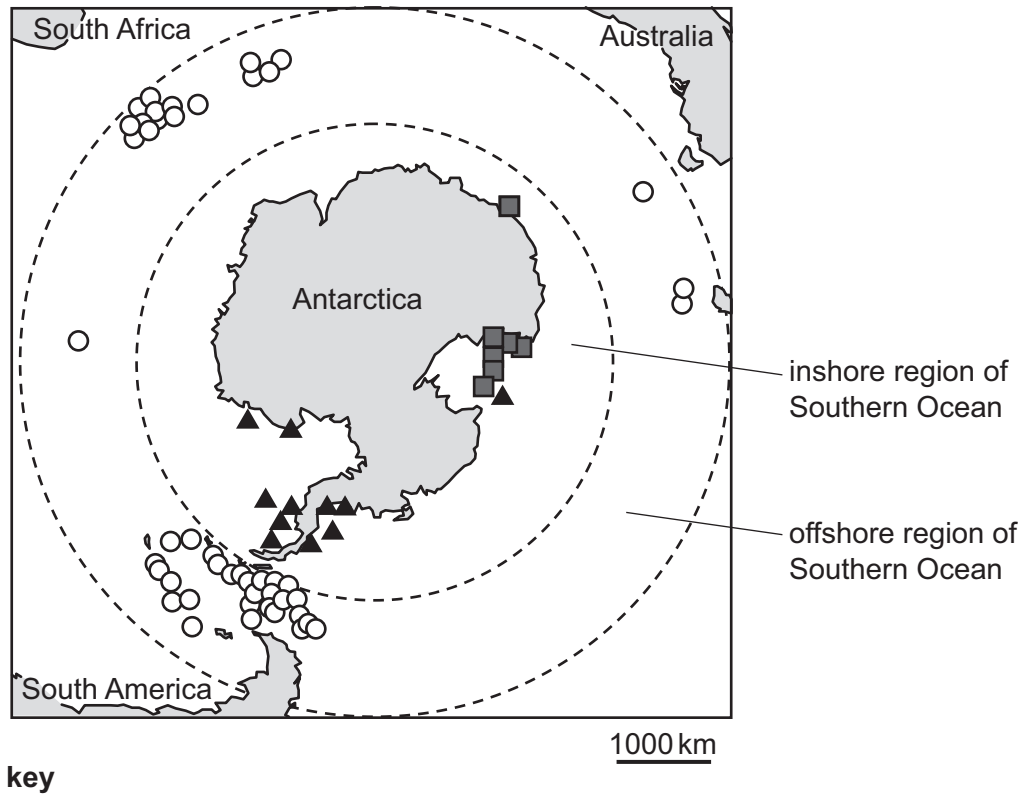


Fig. 2.3

There are phenotypic differences between the different types of orca. Fig. 2.4 shows a diagram of a Type **B** orca, a Type **C** orca and a Type **D** orca.

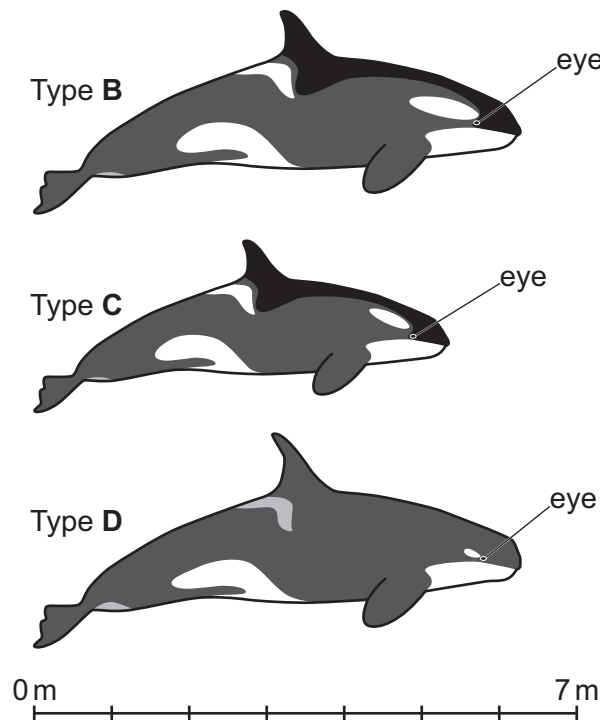


Fig. 2.4

- (i) With reference to Fig. 2.4, state **one** way in which the Type **D** orca is different from both the Type **B** orca and the Type **C** orca.

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 [1]

- (ii) Phenotypic differences between Type **D** orca and the other types of orca shown in Fig. 2.4 could have resulted from the process of genetic drift, including the founder effect.

Suggest how genetic drift could result in phenotypic differences between Type **D** orca and the other types of orca shown in Fig. 2.4.

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 [3]



- (c) In the future, the different types of orca may be classified as separate species. If so, some of these newly classified species will have very small population sizes.

Suggest **two** factors, **other** than population size, that should be monitored when assessing the conservation status of any newly classified species of orca.

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[Total: 10]



Question 3 starts on page 12.